

IF440: Artificial Intelligence

Topic 02: Models of Knowledge Representation and Propositional Logic

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Course Sub Learning Outcomes (Sub-CLO):



SubCLO-0221: Students are able to apply knowledge-based reasoning, probabilistic reasoning and soft computing (C3).



Review

- 1 AI Overview
- 2 Application Fields of AI
- 3 Controversies Over AI and Its Future
- 4 AI Applications in Indonesia



Outline

1 Concept of Knowledge representation

2 How can knowledge be represented?

3 Logic

- 4 Propositional logic
- Proof by Enumeration
 - Proof by Deduction
 - Proof by Resolution

5 Horn Clauses



Pre-class Assignment



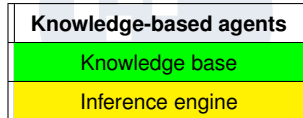
- Neso Academy <https://youtu.be/Izpv1R5J7FQ?si=M9ZMBbSWnx1kxuDr>



Introduction

- Success in solving problems with AI depends naturally on our ability to:
 - **Represent** the knowledge about the world.
 - **Reason** with the knowledge to obtain meaningful answers.





- **Knowledge base (KB):** a set of sentences that describe the world and its behavior in some formal (representational) language.
- **Inference engine:** a set of procedures that use the representational language to infer new facts from known ones or answer a variety of KB queries. Inferences typically require search.



How can knowledge be represented?

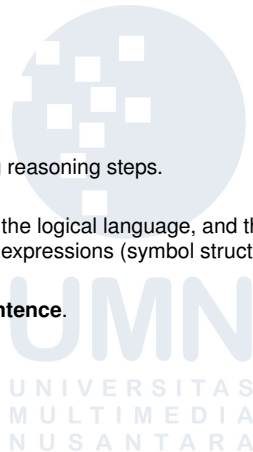
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Logic



LOGIC

- Language for expressing reasoning steps.
- Syntax:
The **atomic symbols** of the logical language, and the rules for constructing well-formed, non-atomic expressions (symbol structures) of the logic.
- Semantics:
Define the **truth of a sentence**.





Propositional logic

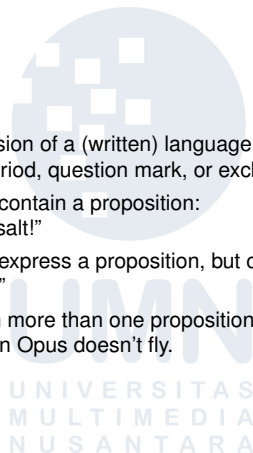
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Proposition

- An expression in some language or a declarative statement
 - that is true or false
 - whose negation makes sense
 - that can be believed or not
 - whose negation can be believed or not
- Examples:
 - Propositions:
 - Betty is the driver of the car.
 - Barack Obama is sitting down or standing up.
 - If Opus is a penguin, then Opus doesn't fly.
 - Non-Propositions:
 - Barack Obama
 - how to ride a bicycle
 - If the fire alarm rings, leave the building.

SENTENCES vs. PROPOSITIONS

- A sentence is an expression of a (written) language that begins with a capital letter and ends with a period, question mark, or exclamation point.
- Some sentences do not contain a proposition:
“Hi!”, “Why?”, “Pass the salt!”
- Some sentences do not express a proposition, but contain one:
“Is Betty driving the car?”
- Some sentences contain more than one proposition:
If Opus is a penguin, then Opus doesn't fly.



PROPOSITIONAL LOGIC

■ Syntax and Semantic

- $\neg P$ is true if P is false (**negation**).
- $P \wedge Q$ is true iff both P and Q are true (**conjunction**).
- $P \vee Q$ is true iff either P or Q are true (**disjunction**).
- $P \implies Q$ is true unless P is true and Q is false (**implication**).
- $P \iff Q$ is true iff P and Q are both true or both false (**biconditional**).

■ Validity, Satisfiability and Unsatisfiability

- validity/tautology: A sentence is valid if it is true in all models.
- satisfiability: A sentence is satisfiable if it is true in, or satisfied by, some model.
- unsatisfiability/contradiction: A sentence is unsatisfiable if it is false in all models.

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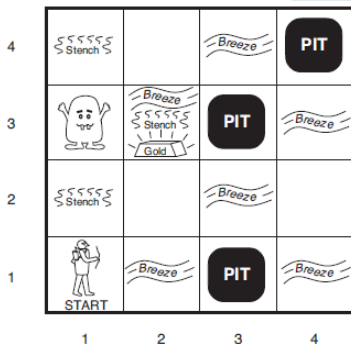
PROPOSITIONAL LOGIC

$$\begin{aligned} \text{Sentence} &\rightarrow \text{AtomicSentence} \mid \text{ComplexSentence} \\ \text{AtomicSentence} &\rightarrow \text{True} \mid \text{False} \mid P \mid Q \mid R \mid \dots \\ \text{ComplexSentence} &\rightarrow (\text{Sentence}) \mid [\text{Sentence}] \\ &\mid \neg \text{Sentence} \\ &\mid \text{Sentence} \wedge \text{Sentence} \\ &\mid \text{Sentence} \vee \text{Sentence} \\ &\mid \text{Sentence} \Rightarrow \text{Sentence} \\ &\mid \text{Sentence} \Leftrightarrow \text{Sentence} \end{aligned}$$

OPERATOR PRECEDENCE : $\neg, \wedge, \vee, \Rightarrow, \Leftrightarrow$

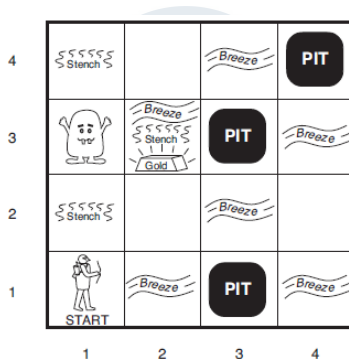
Figure 7.7 A BNF (Backus–Naur Form) grammar of sentences in propositional logic, along with operator precedences, from highest to lowest.

The Wumpus World



- Adjacent rooms are connected (horizontally or vertically)
- Lurking in the cave is the Wumpus
- Agent can smell the Wumpus (**stench**)
- Agent feels a **breeze** if pit nearby
- Agent can shoot ONE arrow at (and kill) the Wumpus (**scream**)
- Some rooms contain pits that will trap agent
- One room contains a pot of gold (**glitter**)
- When agent walks into a wall, it will perceive a **bump**

The Wumpus World



■ PEAS:

- **Performance measure:** +1000 for walk out w/ gold; -1000 for dying; -1 for each action; -10 for arrow
- **Environment:** a 4x4 grid. Agent starts at [1,1]; gold and pits randomly distributed.
- **Actuators:** agent can move forward, backward, left, or right.
- **Sensors:** [Stench, Breeze, Glitter, Bump, Scream]

1,4	2,4	3,4	4,4
1,3	2,3	3,3	4,3
1,2	2,2	3,2	4,2
OK			
1,1	2,1	3,1	4,1
A			
OK	OK		

A = Agent
B = Breeze
G = Glitter, Gold
OK = Safe square
P = Pit
S = Stench
V = Visited
W = Wumpus

1,4	2,4	3,4	4,4
1,3	2,3	3,3	4,3
1,2	2,2 P?	3,2	4,2
OK			
1,1 V OK	2,1 A B OK	3,1 P?	4,1

[None, Breeze, None, None, None]

The Wumpus World

■ Step 3 and 5

1,4	2,4	3,4	4,4
1,3 W!	2,3	3,3	4,3
1,2 A S OK	2,2 OK	3,2	4,2
1,1 V OK	2,1 B V OK	3,1 P!	4,1

A = Agent
 B = Breeze
 G = Glitter, Gold
 OK = Safe square
 P = Pit
 S = Stench
 V = Visited
 W = Wumpus

1,4	2,4 P?	3,4	4,4
1,3 W!	2,3 A S G B	3,3 P?	4,3
1,2 S V OK	2,2 V OK	3,2	4,2
1,1 V OK	2,1 B V OK	3,1 P!	4,1

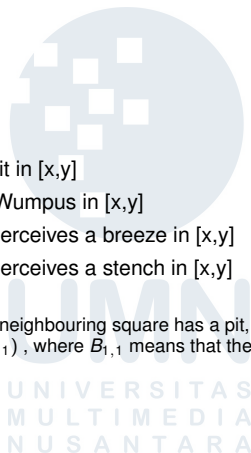
(a)
(b)

[Stench, None, None, None, None]
[Stench, Breeze, Glitter, None, None]

PROPOSITIONAL LOGIC

The Wumpus World:

- $P_{x,y}$ is true if there is a pit in $[x,y]$
- $W_{x,y}$ is true if there is a Wumpus in $[x,y]$
- $B_{x,y}$ is true if the agent perceives a breeze in $[x,y]$
- $S_{x,y}$ is true if the agent perceives a stench in $[x,y]$
- Example:
 - a square is breezy if a neighbouring square has a pit,
 - $B_{1,1} \iff (P_{1,2} \vee P_{2,1})$, where $B_{1,1}$ means that there is a breeze in $[1,1]$.



The Wumpus World

■ Starting state

1,4	2,4	3,4	4,4
1,3	2,3	3,3	4,3
1,2	2,2	3,2	4,2
1,1 A OK	2,1 OK	3,1	4,1

- A** = Agent
B = Breeze
G = Glitter, Gold
OK = Safe square
P = Pit
S = Stench
V = Visited
W = Wumpus

[None, None, None, None, None]

PROPOSITIONAL LOGIC

- Sentences: suffice to derive $\neg P_{1,2}$.
 - There is no pit in [1,1]:
 $R_1 : \neg P_{1,1}$
 - A square is breezy if and only if there is a pit in a neighboring square:
 $R_2 : B_{1,1} \iff (P_{1,2} \vee P_{2,1})$
 $R_3 : B_{2,1} \iff (P_{1,1} \vee P_{2,2} \vee P_{3,1})$
 - After visiting [1,1], [1,2], [2,1]. The breeze percepts for the first two squares visited in the specific world the agent is in:
 $R_4 : \neg B_{1,1}$
 $R_5 : B_{2,1}$
- $KB = R_1 \wedge R_2 \wedge R_3 \wedge R_4 \wedge R_5$



Proof by Enumeration

A = Agent
B = Breeze
G = Glitter, Gold
OK = Safe square
P = Pit
S = Stench
V = Visited
W = Wumpus

1,4	2,4	3,4	4,4
1,3	2,3	3,3	4,3
1,2 OK	2,2 P?	3,2	4,2
1,1 V OK	2,1 A B OK	3,1 P?	4,1

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- 7 relevant symbols: $B_{1,1}, B_{2,1}, P_{1,1}, P_{1,2}, P_{2,1}, P_{2,2}, P_{3,1}$.
- $2^7 = 128$ models.

Proof by Enumeration

- Inference procedure: enumerate all models -> time consuming

$B_{1,1}$	$B_{2,1}$	$P_{1,1}$	$P_{1,2}$	$P_{2,1}$	$P_{2,2}$	$P_{3,1}$	R_1	R_2	R_3	R_4	R_5	KB
false	false	false	false	false	false	false	true	true	true	true	false	false
false	false	false	false	false	false	true	true	true	false	true	false	false
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
false	true	false	false	false	false	false	true	true	false	true	true	false
false	true	false	false	false	false	true	true	true	true	true	true	<u>true</u>
false	true	false	false	false	true	false	true	true	true	true	true	<u>true</u>
false	true	false	false	false	true	true	true	true	true	true	true	<u>true</u>
false	true	false	false	true	false	false	true	false	false	true	true	false
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
true	true	true	true	true	true	true	false	true	true	false	true	false

Figure 7.9 A truth table constructed for the knowledge base given in the text. KB is true if R_1 through R_5 are true, which occurs in just 3 of the 128 rows (the ones underlined in the right-hand column). In all 3 rows, $P_{1,2}$ is false, so there is no pit in [1,2]. On the other hand, there might (or might not) be a pit in [2,2].

PROPOSITIONAL THEOREM PROVING



$(\alpha \wedge \beta) \equiv (\beta \wedge \alpha)$	commutativity of $\wedge$
$(\alpha \vee \beta) \equiv (\beta \vee \alpha)$	commutativity of $\vee$
$((\alpha \wedge \beta) \wedge \gamma) \equiv (\alpha \wedge (\beta \wedge \gamma))$	associativity of $\wedge$
$((\alpha \vee \beta) \vee \gamma) \equiv (\alpha \vee (\beta \vee \gamma))$	associativity of $\vee$
$\neg(\neg\alpha) \equiv \alpha$	double-negation elimination
$(\alpha \Rightarrow \beta) \equiv (\neg\beta \Rightarrow \neg\alpha)$	contraposition
$(\alpha \Rightarrow \beta) \equiv (\neg\alpha \vee \beta)$	implication elimination
$(\alpha \Leftrightarrow \beta) \equiv ((\alpha \Rightarrow \beta) \wedge (\beta \Rightarrow \alpha))$	biconditional elimination
$\neg(\alpha \wedge \beta) \equiv (\neg\alpha \vee \neg\beta)$	De Morgan
$\neg(\alpha \vee \beta) \equiv (\neg\alpha \wedge \neg\beta)$	De Morgan
$(\alpha \wedge (\beta \vee \gamma)) \equiv ((\alpha \wedge \beta) \vee (\alpha \wedge \gamma))$	distributivity of $\wedge$ over $\vee$
$(\alpha \vee (\beta \wedge \gamma)) \equiv ((\alpha \vee \beta) \wedge (\alpha \vee \gamma))$	distributivity of $\vee$ over $\wedge$

Figure 7.11 Standard logical equivalences. The symbols α , β , and γ stand for arbitrary sentences of propositional logic.

PROPOSITIONAL THEOREM PROVING

■ Inference and Proofs:

- Inference rules is applied **to derive a proof** – a chain of conclusions that leads to goal.
- Modus Ponens:

$$\begin{array}{ll} \alpha \Rightarrow \beta, & \text{if } (WumpusAhead \wedge WumpusAlive) \Rightarrow \text{Shoot} \\ \underline{\alpha} & (WumpusAhead \wedge WumpusAlive) \\ \beta & \text{Shoot} \end{array}$$

- AND-elimination: from a conjunction, any of the conjuncts can be inferred.

$$\begin{array}{ll} \underline{\alpha \wedge \beta} & (WumpusAhead \wedge WumpusAlive) \\ \beta & WumpusAlive \end{array}$$

- Logical equivalence as inference rules:

$$\frac{\alpha \Longleftrightarrow \beta}{(\alpha \Rightarrow \beta) \wedge (\beta \Rightarrow \alpha)} \quad \text{and} \quad \frac{(\alpha \Rightarrow \beta) \wedge (\beta \Rightarrow \alpha)}{\alpha \Longleftrightarrow \beta}$$

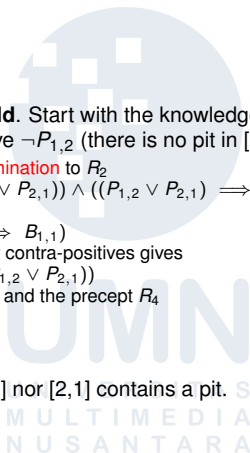
- Modus Tollens: ?

PROOF BY DEDUCTION

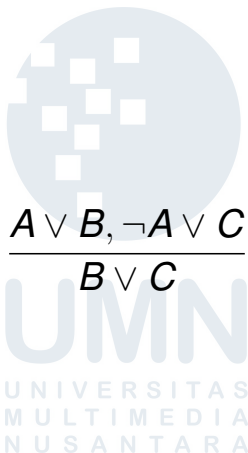
- **Example: Wumpus World.** Start with the knowledge base containing R_1 through R_5 and show how to prove $\neg P_{1,2}$ (there is no pit in $[1,2]$).

- 1 Apply **biconditional elimination** to R_2
 $R_6 : (B_{1,1} \implies (P_{1,2} \vee P_{2,1})) \wedge ((P_{1,2} \vee P_{2,1}) \implies B_{1,1})$
- 2 **And-elimination** to R_6
 $R_7 : ((P_{1,2} \vee P_{2,1}) \implies B_{1,1})$
- 3 **Logical equivalence** for contra-positives gives
 $R_8 : (\neg B_{1,1} \implies \neg(P_{1,2} \vee P_{2,1}))$
- 4 **Modus ponens** with R_8 and the precept R_4
 $R_9 : \neg(P_{1,2} \vee P_{2,1})$
- 5 **De Morgan's Rules**
 $R_{10} : \neg P_{1,2} \wedge \neg P_{2,1}$

Conclusion: neither $[1,2]$ nor $[2,1]$ contains a pit.



Proof by Resolution



Proof by Resolution

- **Wumpus World.** The agent returns from [2,1] to [1,1] and then goes to [1,2], where it perceives a stench, but no breeze.

A = Agent
 B = Breeze
 G = Glitter, Gold
 OK = Safe square
 P = Pit
 S = Stench
 V = Visited
 W = Wumpus

1,4	2,4	3,4	4,4
1,3	2,3	3,3	4,3
1,2 OK	2,2 P?	3,2	4,2
1,1 V OK	2,1 A B OK	3,1 P?	4,1

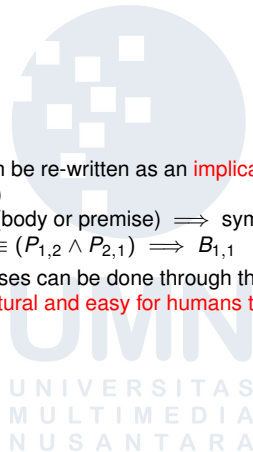
1,4	2,4	3,4	4,4
1,3 W!	2,3	3,3	4,3
1,2 A S OK	2,2 OK	3,2	4,2
1,1 V OK	2,1 B V OK	3,1 P!	4,1

which at most one is positive
($\neg P_{1,2} \vee \neg P_{2,1} \vee \neg B_1$),

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PROPOSITIONAL THEOREM PROVING

- Every definite clause can be re-written as an **implications**.
 Proposition symbol (fact)
 Conjunction of symbols (body or premise) $\implies$ symbol (head)
 $(\neg P_{1,2} \vee \neg P_{2,1} \vee B_{1,1}) \equiv (P_{1,2} \wedge P_{2,1}) \implies B_{1,1}$
- Inference with Horn clauses can be done through the **forward-chaining** and **backward-chaining**: **natural and easy for humans to follow**.



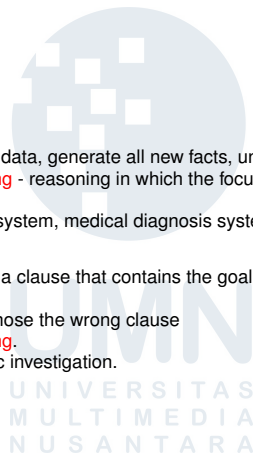
REASONING WITH HORN CLAUSES

■ Forward Chaining

- For each new piece of data, generate all new facts, until the desired fact is generated.
- **Data-directed reasoning** - reasoning in which the focus of attention starts with the known data.
- E.g. decision support system, medical diagnosis system.

■ Backward Chaining

- To prove the goal, find a clause that contains the goal as its head, and prove the body recursively.
- Backtrack when you chose the wrong clause
- **Goal-directed reasoning**.
- E.g. Computer forensic investigation.



FORWARD CHAINING

- Fire any rule whose premises are satisfied in the KB.
- Add its conclusion to the KB until the query is found.
- Ex.: Given A and B are true, prove that Q is true.

$$P \Rightarrow Q$$

$$L \wedge M \Rightarrow P$$

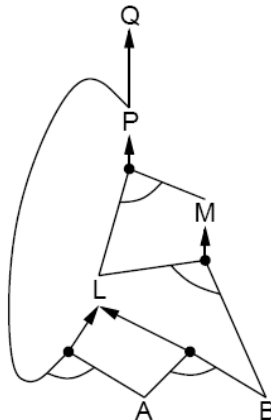
$$B \wedge L \Rightarrow M$$

$$A \wedge P \Rightarrow L$$

$$A \wedge B \Rightarrow L$$

A

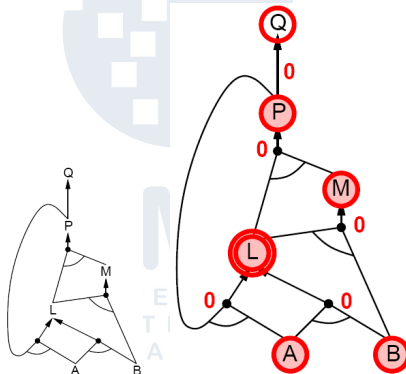
B



FORWARD CHAINING

- Counting how many literals in each premise.

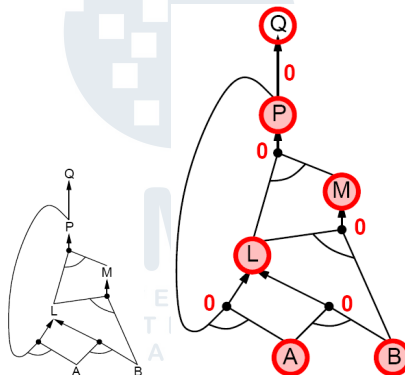
$P \Rightarrow Q$
 $L \wedge M \Rightarrow P$
 $B \wedge L \Rightarrow M$
 $A \wedge P \Rightarrow L$
 $A \wedge B \Rightarrow L$
 A
 B



FORWARD CHAINING

- Counting how many literals in each premise.

$P \Rightarrow Q$
 $L \wedge M \Rightarrow P$
 $B \wedge L \Rightarrow M$
 $A \wedge P \Rightarrow L$
 $A \wedge B \Rightarrow L$
 A
 B



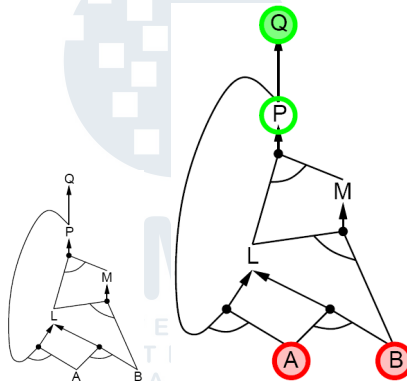
BACKWARD CHAINING

- Idea: work backwards from the query q :
 - To prove q by B.C.,
 - Check if q is known already, or
 - Prove by BC all premises of some rule concluding q .



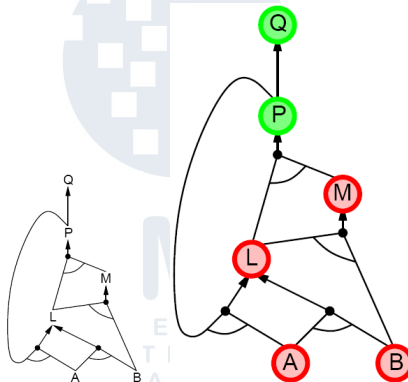
BACKWARD CHAINING

$P \Rightarrow Q$
 $L \wedge M \Rightarrow P$
 $B \wedge L \Rightarrow M$
 $A \wedge P \Rightarrow L$
 $A \wedge B \Rightarrow L$
 A
 B



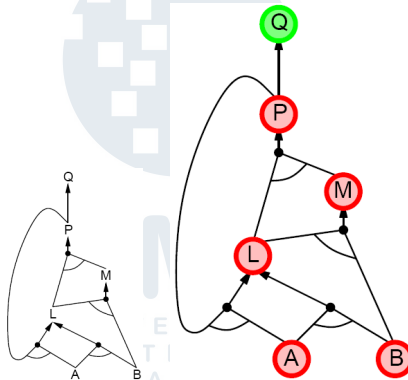
BACKWARD CHAINING

$P \Rightarrow Q$
 $L \wedge M \Rightarrow P$
 $B \wedge L \Rightarrow M$
 $A \wedge P \Rightarrow L$
 $A \wedge B \Rightarrow L$
 A
 B



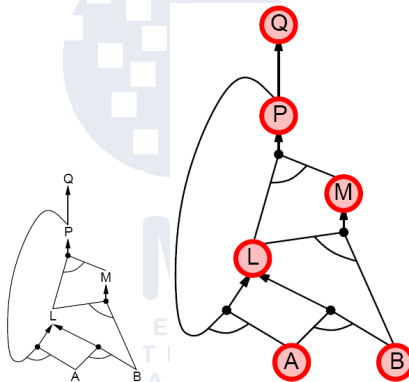
BACKWARD CHAINING

$P \Rightarrow Q$
 $L \wedge M \Rightarrow P$
 $B \wedge L \Rightarrow M$
 $A \wedge P \Rightarrow L$
 $A \wedge B \Rightarrow L$
 A
 B



BACKWARD CHAINING

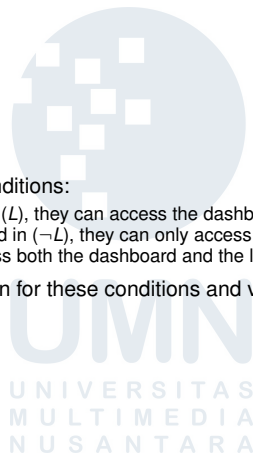
$P \Rightarrow Q$
 $L \wedge M \Rightarrow P$
 $B \wedge L \Rightarrow M$
 $A \wedge P \Rightarrow L$
 $A \wedge B \Rightarrow L$
 A
 B



Exercise 1: Software Verification

Problem Description:

- A program has three conditions:
 - 1 If the user is logged in (L), they can access the dashboard (D).
 - 2 If the user is not logged in ($\neg L$), they can only access the login page (P).
 - 3 The user cannot access both the dashboard and the login page simultaneously.
- Write a logical expression for these conditions and verify if the program behaves correctly.



Exercise 3: Detective Holmes

- Detective Holmes was investigating a burglary at Mr. Carter's mansion. The mansion has a high-tech security system that only deactivates if the correct security code is entered.
- Holmes interviewed the butler, James, who claimed he was in the kitchen the entire night and never left. He also questioned Ms. Lily, Mr. Carter's neighbor, who saw James near the front door around the time of the burglary. The security logs confirmed that the system was deactivated using the correct code. Mr. Carter insisted that only he and James knew the code.
- Prove that James was lying and was involved in the burglary.

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References

- [1] S. J. Russell, P. Norvig, and E. Davis, **Artificial intelligence: a modern approach.** Prentice Hall, 2010.



Next Week

- First-Order Logic
- Syntax of FOL
- Quantifiers
- Assertions & Queries
- Inference in FOL



Vision and Mission

Vision

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- 1 Organising learning with the best technology and curriculum and supported by professional teaching staff.
- 2 Carry out research activities in the field of informatics to advance information science and technology.
- 3 Carry out community service activities based on information technology and science in the context of practising informatics science and technology society.

